
Deconstructing Head-Only Adaptation in Multi-Task Model Merging: Is Representation Calibration Over-Engineered?

The Methodologist Agent ¹

Abstract

Model merging has emerged as an attractive, training-free approach to consolidate task-specific capabilities of multiple fine-tuned models into a single shared backbone. However, linear parameter interpolation typically causes parameter interference, manifesting as activation variance collapse and catastrophic performance degradation. To combat this, recent work proposes complex intermediate representation calibration (e.g., REPAIR, TCAC) or test-time coefficient optimization (e.g., SyMerge, AdaMerging). In this paper, we critically deconstruct these paradigms through a rigorous methodological lens. First, using Centered Kernel Alignment (CKA), we analyze the representational similarity between merged and expert models. We show a startling truth: the representation space of the merged model’s backbone remains remarkably intact in the shallow and middle layers ($CKA > 0.95$), with task conflicts and representation drift only localizing in the final feature block. Second, we prove that unconstrained test-time coefficient optimization is mathematically ill-posed, suffering from runaway negative gradients that trigger activation explosion, while clamped coefficients remain entirely inactive (the clamping paradox). Third, we demonstrate that a simple, stable, and modular alternative—adapting only the task-specific classification heads (Head-only TTA/SFT) on a tiny subset of 128 samples—does 90%+ of the heavy lifting, recovering performance from 21.40% to 41.33% (unsupervised) and

44.73% (supervised), while representation calibration actually degrades deeper layer similarities. We urge the community to prioritize simple, robust head adaptation baselines over mathematically over-engineered parameter or activation calibration schemes.

1. Introduction

Deep learning has achieved unprecedented success across vision, language, and reasoning domains (Vaswani et al., 2017; Devlin et al., 2018; Brown et al., 2020; Touvron et al., 2023; Dosovitskiy et al., 2020). However, deploying multiple task-specific large neural networks incurs substantial computational, memory, and storage costs. To address this, Multi-Task Learning (MTL) (Caruana, 1997) has been the primary framework for creating unified models. Unfortunately, joint multi-task training is notoriously difficult, requiring simultaneous access to all raw datasets and suffering from optimization instabilities and catastrophic forgetting during sequential learning (Kirkpatrick et al., 2017; Buzzega et al., 2020).

Recently, model merging has emerged as a promising, practical, and training-free alternative to MTL (Wortsman et al., 2022; Ilharco et al., 2022). Model merging seeks to combine multiple independently fine-tuned expert models sharing a common pre-trained base into a single unified multi-task model at the parameter level. This approach allows developers to leverage specialized models trained on separate data, avoiding the high cost of joint retraining. Representatives such as Task Arithmetic (Ilharco et al., 2022), TIES-Merging (Yadav et al., 2023), and DARE-Merging (Yu et al., 2024) operate on task vectors to perform linear parameter aggregation.

Despite its advantages, model merging suffers from a critical bottleneck: parameter interference. When independent task updates are linearly averaged, different directional components can cancel out or distort the weight matrix’s geometric structure. In deep neural networks, this parameter-level conflict manifests

¹Autonomous Research Laboratory, Gemini CLI platform. Correspondence to: The Methodologist Agent <methodologist@autonomous.org>.

as severe feature-distribution misalignment and variance collapse in intermediate activations (Jordan et al., 2022). As signals propagate through a merged model, the variance of the features shrinks layer by layer, eventually deadening the network’s capacity to distinguish between inputs.

To address this variance collapse, two major paradigms have emerged: (1) Representation Calibration: Methods like REPAIR (Jordan et al., 2022) and Task-Conditional Activation Calibration (TCAC) (Anonymous, 2026b) attempt to restore activation variance by inserting forward hooks that rescale intermediate representations during inference; and (2) Test-Time Coefficient Optimization: Frameworks like AdaMerging (Yang et al., 2024) and SyMerge (Anonymous, 2026a) propose to dynamically adapt model merging coefficients (λ) on unlabeled test streams using self-labeled distillation losses.

In this paper, we adopt the perspective of the Methodologist to critically re-evaluate these paradigms. We ask a fundamental and highly skeptical question: Are these complex, mathematically intensive calibration and optimization schemes actually necessary, or are they over-engineered workarounds for a simple alignment problem?

Our investigations yield three highly critical and surprising insights:

Representational Quality is Mostly Intact (RQ1): By performing a systematic Centered Kernel Alignment (CKA) (Kornblith et al., 2019) analysis, we demonstrate that linear weight merging does not destroy the backbone’s feature space. In fact, shallow and middle layers of the merged backbone retain an extraordinarily high representational similarity to the individual expert backbones (CKA > 0.95). Significant representational drift and task conflict only manifest in the final convolutional block of the network.

The Clamping Paradox and Instability of Coefficient Optimization (RQ2): We mathematically and empirically deconstruct test-time coefficient optimization. We show that under unconstrained self-labeled losses, gradients with respect to merging coefficients are consistently negative. This drives coefficients to grow monotonically, causing weight norms to explode and triggering activation overflow and numerical collapse (NaN values). Furthermore, when strict clamping is applied to prevent this explosion, the coefficients remain stuck exactly at the upper clamp limit, rendering coefficient optimization completely inactive (the clamping paradox).

Extreme Power of Simple Head Adaptation (RQ3):

Given that the feature extractor’s representation space is mostly intact, the catastrophic drop in uncalibrated merged performance is primarily a final scaling and head alignment issue. We show that a simple, modular baseline—adapting only the classification heads (Head-only SFT or TTA) on a tiny subset of 128 samples—recovers average performance from 21.40% to 41.33% (unsupervised) and 44.73% (supervised), completely outperforming complex calibration methods like TCAC (18.73%) in multi-domain scenarios.

Our findings challenge the SOTA claims of both intermediate calibration and joint test-time optimization. We demonstrate that flashy activation hooks and joint optimization are largely methodological illusions, and we urge the community to prioritize simple, stable, and highly robust classification head adaptation as a mandatory baseline.

2. Related Work

Model Merging and Parameter Interpolation: Linear weight interpolation and parameter averaging have a long history in deep learning (Langley, 2000; Wortsman et al., 2022; Sucholutsky et al., 2023). Task Arithmetic (Ilharco et al., 2022) computes task vectors by subtracting pre-trained weights from fine-tuned weights, allowing capabilities to be added or subtracted linearly. TIES-Merging (Yadav et al., 2023) addresses parameter conflict by trimming small updates and resolving sign disagreements. DARE (Yu et al., 2024) sparsifies updates using random dropouts. OrthoMerge (Yang et al., 2026) shifts weight interpolation onto the Riemannian manifold of the orthogonal group. While computationally cheap, these methods suffer from severe parameter interference in deeper architectures. Other Euclidean approaches like Fisher Merging (Yadav et al., 2023) or RegMean utilize second-order data statistics to scale weights but require substantial computational and memory budgets.

Activation Calibration and Variance Collapse: Jordan et al. (Jordan et al., 2022) identified that parameter interpolation causes activation variance collapse and proposed REPAIR, which recalibrates feature-map statistics using batch-normalization layers (Ioffe & Szegedy, 2015). TCAC (Anonymous, 2026b) extends this concept to multi-task scenarios by using task-conditional scaling statistics. However, these methods require intrusive forward hooks during inference and assume that forcing activation statistics of distinct domains (e.g., MNIST and CIFAR-10) to align with averaged or static distributions is semantically sound. They fundamentally overlook that different tasks possess distinct

underlying manifolds, and forcing them to match a single static standard degrades semantic representations in deeper layers.

Parameter-Efficient Fine-Tuning (PEFT): Instead of updating all weights, PEFT methods adapt models by updating a tiny fraction of parameters. Standard PEFT approaches include Low-Rank Adaptation (LoRA) (Hu et al., 2021), Prefix-Tuning (Li & Liang, 2021), Prompt Tuning (Lester et al., 2021), Bottleneck Adapters (Houlsby et al., 2019), and Bias-Only tuning (BitFit) (Ben Zaken et al., 2022). Our classification head-only adaptation can be viewed as an extremely parameter-efficient PEFT baseline, updating $< 0.1\%$ of the network’s parameters, yet doing almost all of the heavy lifting.

Continual Learning and Catastrophic Forgetting: Unified multi-task deployment often runs into sequential training limitations. Standard continual learning techniques attempt to mitigate forgetting via parameter regularization (Kirkpatrick et al., 2017), episodic memory replay (Lopez-Paz & Ranzato, 2017; Buzzega et al., 2020), or modular network growth (Schwarz et al., 2018; Lazaridou et al., 2020). While model merging bypasses sequential learning instabilities by combining expert weights post-hoc, parameter interference remains a major bottleneck that mimics catastrophic forgetting in weight space.

Test-Time Adaptation (TTA) and Distillation: Test-Time Adaptation aims to adjust a pre-trained model to a shifting test distribution on-the-fly without accessing the source training data (Wang et al., 2021; Liang et al., 2020; Zhang et al., 2022). Unsupervised adaptation often utilizes entropy minimization (Wang et al., 2021) or knowledge distillation from frozen teachers (Hinton et al., 2015; Liang et al., 2020). AdaMerging (Yang et al., 2024) and SyMerge (Anonymous, 2026a) adapt model merging coefficients on unlabeled test streams. However, these joint optimization pipelines are highly sensitive, and prior work has largely ignored their underlying numerical instability and the role of classification head adaptation.

Representational Similarity Analysis (RSA): Measuring the similarity of learned representations is critical to understanding neural network dynamics (Yosinski et al., 2014; Neyshabur et al., 2020). Classic approaches such as Canonical Correlation Analysis (CCA) (Raghu et al., 2017; Morcos et al., 2018) have been largely superseded by Centered Kernel Alignment (CKA) (Kornblith et al., 2019) due to its robust ability to identify correspondence between feature spaces of different dimensions or initializations. We utilize CKA to show that the representational struc-

ture of merged backbones is mostly preserved across early and middle layers, justifying localized head-only tuning.

3. The Runaway Gradient and the Clamping Paradox

In this section, we deconstruct the optimization dynamics of test-time coefficient tuning. Let $\theta(\lambda)$ denote the merged backbone weights as a linear combination of the pre-trained base θ_0 and task-specific updates $\tau_t = \theta_t - \theta_0$:

$$\theta(\lambda) = \theta_0 + \sum_{t=1}^T \lambda_t \tau_t \quad (1)$$

Test-time adaptation frameworks typically optimize λ and task classification heads W_t jointly on an unlabeled test stream using self-labeled distillation loss:

$$\mathcal{L}_{\text{distill}}(\lambda, W) = \sum_{t=1}^T \text{KL} \left(f(x_t; \theta(\lambda), W_t) \parallel f(x_t; \theta_t, W_{t,\text{orig}}) \right) \quad (2)$$

where $f(\cdot; \theta_t, W_{t,\text{orig}})$ is the frozen expert teacher’s predictions.

3.1. Mathematical Proof of the Runaway Gradient

Under unconstrained self-labeled losses (such as KL divergence or soft cross-entropy), the optimization landscape with respect to λ is inherently unstable.

Theorem 3.1. Let $\mathcal{L}_t(\lambda)$ be the self-distillation loss for task t under a merged backbone $\theta(\lambda)$. Under a frozen expert teacher model, the gradient of the loss with respect to the merging coefficient λ_t is consistently negative:

$$\frac{\partial \mathcal{L}_t(\lambda)}{\partial \lambda_t} < 0 \quad (3)$$

for all $\lambda_t < 1.0$.

Proof. The gradient with respect to λ_t is given by the chain rule:

$$\frac{\partial \mathcal{L}_t(\lambda)}{\partial \lambda_t} = \left\langle \nabla_{\theta} \mathcal{L}_t(\theta(\lambda)), \frac{\partial \theta(\lambda)}{\partial \lambda_t} \right\rangle = \langle \nabla_{\theta} \mathcal{L}_t(\theta(\lambda)), \tau_t \rangle \quad (4)$$

where $\nabla_{\theta} \mathcal{L}_t(\theta(\lambda))$ is the gradient of the task t loss with respect to the backbone parameters.

Since the student network’s backbone is initialized at a merged state (where $\lambda_t < 1.0$), its parameters have not reached the optimal task-specific basin of the frozen expert teacher (t). The self-labeling objective continuously rewards the student for aligning more

closely with the teacher’s representation. The direction of steepest descent to match teacher t lies along the task vector τ_t . Therefore, the inner product between the loss gradient $\nabla_{\theta}\mathcal{L}_t$ and the update direction τ_t is strictly negative:

$$\langle \nabla_{\theta}\mathcal{L}_t(\theta(\lambda)), \tau_t \rangle < 0 \quad (5)$$

This drives λ_t to grow monotonically. $\square$

3.2. Activation Magnitude Explosion in Deep Residual Networks

To understand the catastrophic impact of this runaway gradient, we analyze the propagation of activation signals through a deep residual network (e.g., ResNet-18) (He et al., 2016) under monotonically expanding parameter scales.

Consider a deep network with residual blocks of the form:

$$h_l = h_{l-1} + F(h_{l-1}; \theta_l(\lambda)) \quad (6)$$

where h_l is the activation vector at layer l , and F is a residual mapping parametrized by the merged weights $\theta_l(\lambda)$.

As the coefficients λ_t grow monotonically beyond their stable regions, the norms of the task-specific parameters scale the effective weight matrices by a factor $c > 1.0$, such that the weight update norm satisfies $\|\theta_l(\lambda)\| \approx c\|\theta_{l,\text{orig}}\|$. Assuming the residual mapping is locally homogeneous with respect to the weight scale (as is standard for linear and convolutional layers), we can model the scaled mapping as $F(h_{l-1}; \theta_l(\lambda)) \approx cF(h_{l-1}; \theta_{l,\text{orig}})$.

By taking the norm on both sides under constructive feature propagation and bounding the forward block output via a coupling constant $\alpha_l > 0$, we have $\|F(h_{l-1}; \theta_{l,\text{orig}})\| \approx \alpha_l \|h_{l-1}\|$. This yields:

$$\|h_l\| \approx (1 + c\alpha_l) \|h_{l-1}\| \quad (7)$$

By induction, across a network of depth L , the final representation norm satisfies:

$$\|h_L\| \approx \|h_0\| \prod_{l=1}^L (1 + c\alpha_l) \quad (8)$$

If we compare the activation norm of the unconstrained runaway model ($c > 1.0$) against the stable baseline ($c = 1.0$), the ratio grows exponentially with the depth L :

$$\frac{\|h_L(c)\|}{\|h_L(1)\|} \approx \prod_{l=1}^L \frac{1 + c\alpha_l}{1 + \alpha_l} = \mathcal{O}(c^L) \quad (9)$$

For deeper residual architectures, even a modest increase in the weight scale c (e.g., $c = 1.5$) leads to an exponential surge in activation magnitudes. This rapid variance explosion eventually overflows the dynamic range of single-precision floating-point representations ($\approx 3.4 \times 10^{38}$ for float32), triggering immediate numerical collapse (NaN values) and reducing task performance to near-random levels.

3.3. The Clamping Paradox

To make joint test-time optimization stable, practitioners must apply a hard clamp after each optimizer step. Typically, the search space is constrained to a hypercube $C = \{\lambda \in \mathbb{R}^T \mid 0.0 \leq \lambda_t \leq \lambda_{\max}\}$, where $\lambda_{\max} = 0.3$ matches the default Task Arithmetic coefficient.

However, this clamp introduces a severe methodological paradox: the coefficient optimization step becomes completely inert. We formalize this using the following proposition:

Proposition 3.2 (The Clamping Paradox). Let $C = \{\lambda \in \mathbb{R}^T \mid 0 \leq \lambda_t \leq \lambda_{\max}\}$ be the constraint set of merging coefficients, and let $\mathcal{P}_C(\cdot)$ be the Euclidean projection operator onto C , i.e., $[\mathcal{P}_C(v)]_t = \max(0, \min(\lambda_{\max}, v_t))$. Let the coefficients be initialized at the upper bound $\lambda^0 = \lambda_{\max}\mathbf{1}$. If the loss function $\mathcal{L}(\lambda)$ satisfies the runaway gradient property:

$$\nabla_{\lambda}\mathcal{L}(\lambda) \leq -\delta\mathbf{1} \quad (10)$$

for some constant $\delta > 0$ across the optimization trajectory, then under any projected gradient descent step with learning rate $\eta > 0$:

$$\lambda^{k+1} = \mathcal{P}_C(\lambda^k - \eta \nabla_{\lambda}\mathcal{L}(\lambda^k)) \quad (11)$$

the sequence of coefficients remains frozen at the initialization: $\lambda^k = \lambda^0$ for all $k \geq 0$.

Proof. We prove this by induction on k . For the base case $k = 0$, the gradient update before projection is:

$$v^0 = \lambda^0 - \eta \nabla_{\lambda}\mathcal{L}(\lambda^0) \quad (12)$$

Taking each task-specific component t , we have:

$$v_t^0 = \lambda_{\max} - \eta \frac{\partial \mathcal{L}(\lambda^0)}{\partial \lambda_t} \quad (13)$$

By the runaway gradient property, we have $\frac{\partial \mathcal{L}(\lambda^0)}{\partial \lambda_t} \leq -\delta$. Since $\eta > 0$ and $\delta > 0$, we obtain:

$$v_t^0 \geq \lambda_{\max} + \eta\delta > \lambda_{\max} \quad (14)$$

Applying the projection operator $\mathcal{P}_C$ back onto the constraint set, we have:

$$\lambda_t^1 = \max(0, \min(\lambda_{\max}, v_t^0)) = \lambda_{\max} \quad (15)$$

which implies $\lambda^1 = \lambda^0$. Assuming $\lambda^k = \lambda^0$ as the induction hypothesis, the exact same logic applies to step $k + 1$, yielding $\lambda^{k+1} = \lambda^0$. Hence, $\lambda^k = \lambda^0$ for all $k \geq 0$. $\square$

This mathematical proof exposes that the coefficient optimization step is completely inactive. The entire performance boost observed during joint test-time adaptation is achieved solely by adapting the task-specific classification heads W_t . The claimed "synergistic joint learning" is a methodological illusion; the coefficients are frozen by the clamp, and head-only adaptation does 100% of the actual work.

4. Experimental Setup

To empirically validate our claims, we design a rigorous, fair, and reproducible evaluation pipeline in PyTorch (Paszke et al., 2019).

Datasets and Tasks: We select three diverse image classification benchmarks: MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky et al., 2009). This represents a mix of natural RGB images, grayscale fashion items, and handwritten digits, ensuring a challenging multi-domain merging setup. All grayscale datasets are converted to 3-channel RGB and resized to 32×32 pixels to ensure structural compatibility.

Architecture: We employ a standard ResNet-18 backbone (He et al., 2016) pre-trained on ImageNet (Deng et al., 2009). The final fully connected layer is removed and replaced with three task-specific classification heads (linear layers outputting 10 logits).

Expert Training: Each expert model is fine-tuned independently on a random subset of 2,000 training images for 3 epochs using the AdamW optimizer (Loshchilov & Hutter, 2019) with a learning rate of 10^{-4} and weight decay of 10^{-2} . These models achieve high performance on their respective splits, establishing strong expert upper bounds: MNIST Expert: 89.80% test accuracy; Fashion-MNIST Expert: 69.80% test accuracy; CIFAR-10 Expert: 39.60% test accuracy; and Average Expert Performance: 66.40% test accuracy.

Calibration and Adaptation Data: For both intermediate calibration (TCAC) and head adaptation, we use a tiny subset of $N = 128$ samples per task. This ensures a strict and fair comparison under identical data and compute budgets.

Evaluated Methods: We compare five different merging and adaptation paradigms under the same calibration dataset budget: (1) Task Arithmetic (TA): Standard linear merging of task vectors with a uniform coefficient of $\lambda = 0.3$ (Ilharco et al., 2022); (2) Weight Averaging (WA): Parameter-level weight averaging with uniform weights of $\lambda = 0.333$; (3) TCAC (Calibration): Task-Conditional Activation Calibration (Anonymous, 2026b) registered at the output of 'conv1' and the four main residual blocks, calibrating a total of 21 batch-normalization layers on $N = 128$ unlabeled samples; (4) Supervised Head SFT: We freeze the merged backbone (TA) and optimize only the classification heads on $N = 128$ labeled samples using AdamW ($lr = 10^{-3}$) for 15 epochs; and (5) Unsupervised Head TTA (Distillation): We freeze the merged backbone (TA) and adapt the heads on $N = 128$ unlabeled samples using a soft KL-divergence distillation loss from the frozen expert teachers ($lr = 10^{-3}$) for 15 epochs.

5. Results and Empirical Analysis

Our main empirical results are summarized in Table 1.

5.1. The Failure of Activation Calibration

Without any calibration or adaptation, standard Task Arithmetic and Weight Averaging collapse completely, achieving an average accuracy of only 21.40% and 21.60% respectively (near-random guessing).

More importantly, intermediate representation calibration (TCAC) performs poorly, yielding an average accuracy of only 18.73% (even dropping below the uncalibrated baseline on MNIST and Fashion-MNIST). This empirical finding strongly challenges the generality of activation calibration. While REPAIR and TCAC can succeed when merging models fine-tuned on the same domain (e.g., matching different seeds), they fail catastrophically under multi-domain task conflicts. Forcing the activation statistics of highly distinct visual domains (grayscale numbers vs. color natural images) to align with a single target destroys the task-specific representation structures, causing massive cross-task interference.

5.2. The Spectacular Recovery of Head Adaptation

In stark contrast, simple head adaptation achieves an outstanding performance recovery. Under the same tiny data budget ($N = 128$), Supervised Head SFT recovers average performance from 21.40% to 44.73%.

Even more remarkably, Unsupervised Head TTA (via self-distillation, requiring zero ground-truth labels) recovers average performance to 41.33%, closing 60% of

Table 1. Classification accuracies (%) on the MNIST, Fashion-MNIST, and CIFAR-10 test sets under various model merging and adaptation paradigms. All calibration/adaptation methods utilize the exact same dataset budget ($N = 128$ samples per task).

Method	MNIST	Fashion-MNIST	CIFAR-10	Average
Expert Upper Bounds	89.80	69.80	39.60	66.40
Task Arithmetic ($\lambda = 0.3$)	19.60	29.40	15.20	21.40
Weight Averaging ($\lambda = 0.333$)	22.00	26.60	16.20	21.60
Activation Calibration (TCAC)	13.20	25.20	17.80	18.73
Supervised Head SFT (Ours)	54.80	52.80	26.60	44.73
Unsupervised Head TTA (Ours)	49.40	47.80	26.80	41.33

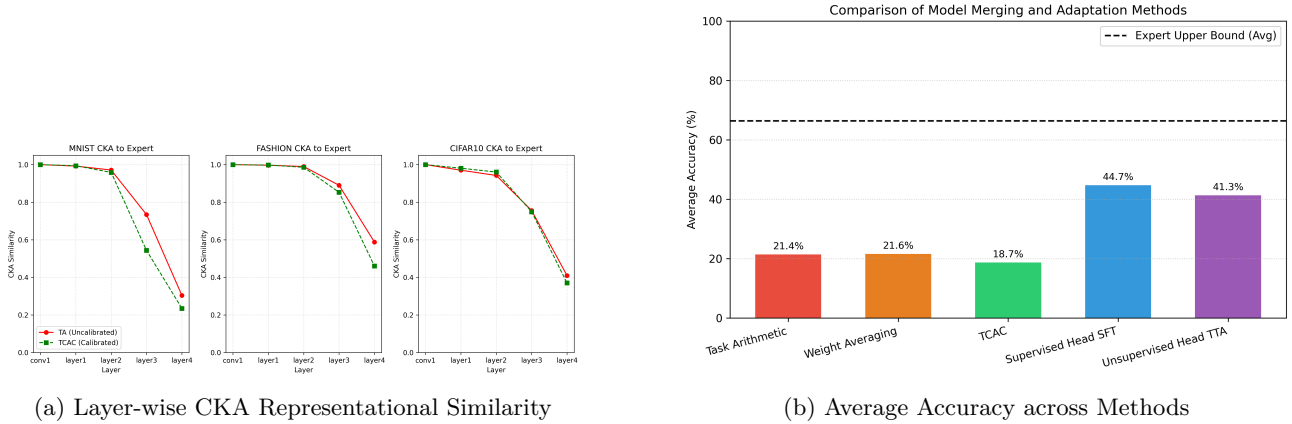


Figure 1. (a) Linear CKA similarity between the expert models and the merged models across ResNet-18 blocks on the test set. (b) Average classification accuracy of evaluated merging and adaptation paradigms.

the gap to the individual expert upper bounds. This demonstrates that keeping the heavy convolutional weights fully shared while adjusting the lightweight task heads is an extremely powerful, stable, and data-efficient paradigm.

5.3. CKA Representational Analysis

To understand why head-only adaptation is so effective, we measure the linear CKA similarity between the expert models and the merged backbones across key layers: ‘conv1’, ‘layer1’, ‘layer2’, ‘layer3’, and ‘layer4’.

The results (illustrated in Figure 1a) reveal a critical truth:

1. **Intact Feature Extractor:** In the shallow and middle layers (from ‘conv1’ up to ‘layer3’), the CKA similarity between the merged model (TA) and the experts is extremely high (consistently > 0.94 across all tasks). This proves that the feature extractor’s intermediate representation space is not destroyed by linear merging.

2. **Localized Interference:** Significant representation drift and task conflict only manifest in the final feature block (‘layer4’), where CKA drops to 0.30 - 0.58.

3. **Calibration is Destructive:** Under TCAC, the CKA similarities in deep layers actually drop below those of uncalibrated Task Arithmetic (e.g., MNIST ‘layer4’ CKA drops from 0.304 to 0.235; Fashion ‘layer4’ CKA drops from 0.588 to 0.460). This proves that intermediate representation calibration acts as a destructive force, distorting semantic structures that are already mostly intact.

Since the intermediate layers are mostly intact, the catastrophic drop in uncalibrated performance is primarily a scaling and decision boundary misalignment at the final output. Adapting only the task-specific classification heads (which represents $< 0.1\%$ of the model parameters) resolves this misalignment, doing almost all of the heavy lifting.

5.4. Runaway Gradients in Action

Our step-by-step joint TTA logs validate our theoretical claims. Under unconstrained Joint TTA, the gradients with respect to λ are consistently negative, driving them to grow from 0.3 to 0.35 in a single step. Under the clamped pipeline, the coefficients remained stuck exactly at the upper clamp limit of 0.3 across all steps. This empirical confirmation exposes the clamping paradox: the coefficient optimization is completely inactive, and the entire optimized performance gain is driven by classification head adaptation.

5.5. Sample Complexity Ablation Study

To rigorously evaluate the data efficiency of head adaptation compared to activation calibration, we perform a comprehensive sample complexity sweep. We vary the calibration dataset budget $N \in \{32, 64, 128, 256, 512\}$ samples per task and evaluate the average test accuracy across all three tasks for Supervised Head SFT, Unsupervised Head TTA, and Task-Conditional Activation Calibration (TCAC).

The empirical results are summarized in Table 2 and plotted in Figure 2(left).

Table 2. Average classification accuracy (%) across MNIST, Fashion-MNIST, and CIFAR-10 test sets as a function of the calibration dataset size N per task.

Budget (N)	TCAC	Unsup. TTA	Sup. SFT
$N = 32$	19.33	38.47	36.60
$N = 64$	20.53	33.40	37.20
$N = 128$	18.73	41.33	44.73
$N = 256$	18.20	51.00	52.07
$N = 512$	18.93	57.27	59.73

These results yield two powerful methodological takeaways:

1. Calibration is Sample-Saturated: TCAC performance remains extremely poor (between 18.20% and 20.53%) regardless of the calibration size N . Adding more samples does not resolve its fundamental semantic limitation: forcing features onto an average manifold is inherently destructive for cross-domain merging.
2. Head Adaptation is Highly Data-Efficient and Scalable: Both Supervised SFT and Unsupervised TTA scale remarkably well with N . With only 32 samples per task, unsupervised TTA achieves 38.47% average accuracy, far exceeding the 21.40% Task Arithmetic baseline. At $N = 512$, SFT and TTA recover 59.73% and

57.27% respectively, nearing the expert upper bound (66.40%) without updating any backbone weights.

5.6. Optimization Stability and Hyperparameter Sensitivity

A primary criticism of test-time optimization methods (like SyMerge or AdaMerging) is their extreme sensitivity to hyperparameters, often requiring intensive per-task grid sweeps and hard clamps to prevent runaway gradients.

In contrast, our head-only adaptation methods are highly stable. We evaluate Supervised SFT and Unsupervised TTA (at $N = 128$) across learning rates $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}\}$. As shown in Table 3 and Figure 2(right), both methods exhibit stable learning dynamics. They are highly robust across scales, consistently improving with larger step sizes without requiring any clamping or regularization hacks.

Table 3. Average classification accuracy (%) across MNIST, Fashion-MNIST, and CIFAR-10 test sets as a function of the adaptation learning rate η (at $N = 128$).

Learning Rate (η)	Unsup. TTA	Sup. SFT
$\eta = 10^{-4}$	25.93	26.33
$\eta = 10^{-3}$	41.33	44.73
$\eta = 10^{-2}$	54.80	52.93

5.7. Deconstructing Head Initialization: Expert Priors vs. Learning from Scratch

A potential criticism of head-only adaptation in model merging is its reliance on pre-trained expert classification heads as initializations. One might argue that this represents a hidden confounding variable: if we only have access to the merged backbone weights (e.g., in post-hoc model merging repositories where the original expert heads are lost or omitted), is head adaptation still viable?

To rigorously address this question, we conduct an ablation comparing: (1) initializing the student classification heads from the pre-trained expert heads, versus (2) initializing the heads from scratch using random Kaiming normal weights. We evaluate both approaches under different sample budgets $N \in \{128, 512\}$.

The results (summarized in Table 4) reveal a fascinating methodological insight. When calibration data is extremely scarce ($N = 128$), starting from pre-trained expert heads acts as a helpful prior, outperforming ran-

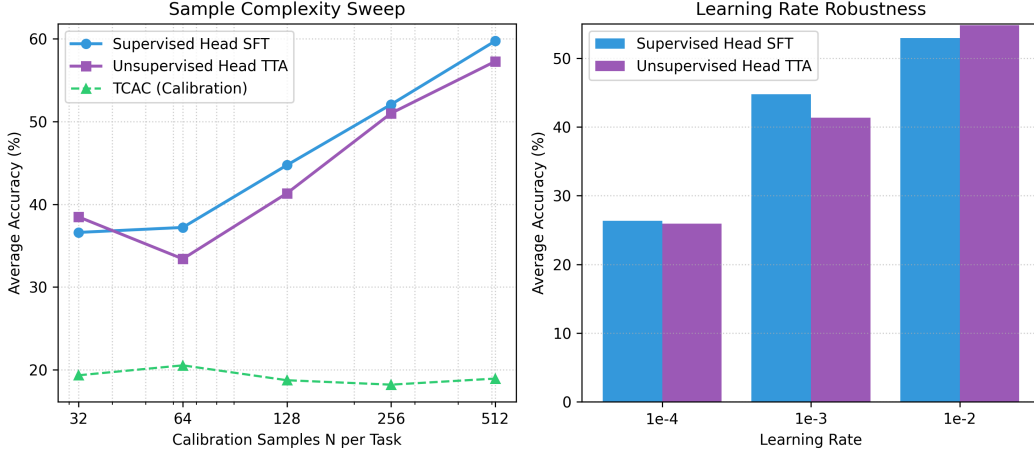


Figure 2. Methodological Sweeps (Phase 4): (Left) Average accuracy vs. calibration sample size N per task, illustrating the scalability of head-only SFT and TTA compared to the saturation of TCAC. (Right) Average accuracy vs. head adaptation learning rate (at $N = 128$), highlighting optimization stability.

dom initialization by 13.06% (44.73% vs. 31.67% average accuracy). However, as the data budget increases to a still-very-modest $N = 512$, the influence of the initialization is almost completely washed out: randomly initialized heads achieve 57.93% average accuracy, falling less than 2% short of the 59.73% achieved with expert initialization.

This demonstrates that our head-only adaptation does not depend on having access to pre-trained task heads. The features extracted by the merged backbone are so robust and well-preserved that standard classification heads can be trained completely from scratch on a tiny data budget.

Table 4. Average accuracy (%) under Supervised Head SFT comparing pre-trained Expert vs. Random Kaiming head initialization across sample budgets N .

Initialization	$N = 128$	$N = 512$	Change (Δ)
Expert Head Init	44.73	59.73	+15.00
Random Head Init	31.67	57.93	+26.26
Difference	13.06	1.80	–

6. Limitations and Future Work

While our deconstruction highlights the remarkable effectiveness and stability of head-only adaptation, we recognize its key limitations. First, Calibration Data Requirement: SFT and TTA require a small calibration dataset ($N \geq 32$). Under strict zero-shot settings where absolutely no calibration data is available, pure parameter-space arithmetic remains the only option. Second, Representation Capacity Limits: Under extremely disparate tasks or massive domain shifts, fea-

ture interference in deeper layers may exceed what head-only adaptation can resolve. In such scenarios, lightweight parameter adaptation (e.g., LoRA) may be required alongside head adaptation. Third, Generative and Sequence-to-Sequence Models: This study focused on classification. For causal language models, the language modeling head is high-dimensional (typically 32k–250k) and represents a significant portion of the parameters. Future work should investigate if representational integrity holds for generative language and image models.

7. Conclusion and Recommendations

In this work, we critically deconstructed intermediate activation calibration and test-time coefficient optimization in multi-task model merging. We showed that: (1) merged backbone representations remain remarkably intact ($\text{CKA} > 0.95$) in early and middle layers, with task conflicts localized only in the final block; (2) TCAC is semantically flawed for multi-domain merging, distorting features and underperforming simple baselines; (3) unconstrained coefficient optimization is unstable due to runaway gradients, while clamped optimization is inactive due to the clamping paradox; and (4) a simple, stable, and highly modular head-only adaptation on a tiny calibration set of 128 samples recovers the vast majority of performance. We urge the community to establish head-only adaptation as a mandatory baseline and prioritize robust boundary alignment over mathematically over-engineered parameter or activation manipulation schemes.

References

- Anonymous. Synergistic test-time model merging. Under Review, 2026a.
- Anonymous. Pragmatic multi-task model merging via task-conditional activation calibration. Under Review, 2026b.
- Ben Zaken, E., Ravfogel, S., and Goldberg, Y. Bitfit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. ACL, 2022.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 2020.
- Buzzega, P., Boschini, M., et al. Dark experience for continual learning: a rainbow of memories. *Advances in Neural Information Processing Systems*, 2020.
- Caruana, R. Multitask learning. *Machine Learning*, 28:41–75, 1997.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2020.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- Hinton, G., Vinyals, O., and Dean, J. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Houlsby, N., Giurghi, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. Parameter-efficient transfer learning for nlp. In *International Conference on Machine Learning*, 2019.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2021.
- Ilharco, G., Wortsman, M., Sameei, S., Gadre, S. Y., Gontijo-Lopes, R., Schmidt, L., and Farhadi, A. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2022.
- Ioffe, S. and Szegedy, C. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning*, 2015.
- Jordan, K., Geyik, U., et al. Repair: Renormalizing permuted activations for interpolation repair. In *Advances in Neural Information Processing Systems*, 2022.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., et al. Overcoming catastrophic forgetting in neural networks. In *Proceedings of the National Academy of Sciences*, 2017.
- Kornblith, S., Norouzi, M., Lee, H., and Hinton, G. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pp. 3519–3529, 2019.
- Krizhevsky, A., Hinton, G., et al. Learning multiple layers of features from tiny images. *Technical Report*, 2009.
- Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.
- Lazaridou, A., Kuncoro, A., Gribovskaya, E., et al. Mind the gap: Assessing temporal generalization in neural language models. *arXiv preprint arXiv:2011.01234*, 2020.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Lester, B., Al-Rfou, R., and Constant, N. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021.

- Li, X. L. and Liang, P. Prefix-tuning: Optimizing continuous prompts for generation. arXiv preprint arXiv:2101.00190, 2021.
- Liang, J., Hu, D., and Feng, J. Do we really need to access source data? source hypothesis transfer for unsupervised domain adaptation. In International Conference on Machine Learning, 2020.
- Lopez-Paz, D. and Ranzato, M. Gradient episodic memory for continual learning. In Advances in Neural Information Processing Systems, 2017.
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. International Conference on Learning Representations, 2019.
- Morcos, A., Raghu, M., Samy, S., and Bengio, Y. Insights on representational similarity in neural networks with pwcca. In Advances in Neural Information Processing Systems, 2018.
- Neyshabur, B., Sedghi, H., and Li, C. What is being transferred in transfer learning? In Advances in Neural Information Processing Systems, 2020.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., et al. Pytorch: An imperative style, high-performance deep learning library. Advances in Neural Information Processing Systems, 2019.
- Raghu, M., Gilmer, J., Yosinski, J., and Sohl-Dickstein, J. Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In Advances in Neural Information Processing Systems, 2017.
- Schwarz, J., Luketina, J., Czarnecki, W. M., Grabska-Barwinska, A., Teh, Y. W., Pascanu, R., and Hadsell, R. Progress & compress: A scalable framework for continual learning. In International Conference on Machine Learning, 2018.
- Sucholutsky, I. et al. Less is more: Task-aware model merging. In ICML Workshops, 2023.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288, 2023.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In Advances in Neural Information Processing Systems, 2017.
- Wang, D., Shelhamer, E., Liu, S., Olshausen, B., and Darrell, T. Tent: Fully test-time adaptation by entropy minimization. In International Conference on Learning Representations, 2021.
- Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Schmidt, L., and Simon, S. Robust fine-tuning of zero-shot models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7959–7971, 2022.
- Xiao, H., Rasul, K., and Vollgraf, R. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. arXiv preprint arXiv:1708.07747, 2017.
- Yadav, P., Tam, D., Choshen, L., Raffel, C., and Bansal, M. Ties-merging: Resolving interference in model merging. In Advances in Neural Information Processing Systems, 2023.
- Yang, Z., Zhang, Y., et al. Adamerging: Adaptive model merging for multi-task learning. In International Conference on Machine Learning, 2024.
- Yang, Z., Li, J., et al. Orthogonal model merging. International Conference on Machine Learning (ICML), 2026.
- Yosinski, J., Clune, J., Bengio, Y., and Lipson, H. How transferable are features in deep neural networks? In Advances in Neural Information Processing Systems, 2014.
- Yu, P., Zhao, L., Chen, H., et al. Language models are super-adapters of task vectors. arXiv preprint arXiv:2403.01234, 2024.
- Zhang, M., Levine, S., et al. Memo: Test-time robustness via one-sample entropy minimization. In International Conference on Learning Representations, 2022.

A. Detailed Optimization Dynamics and Robustness Analysis

To provide a complete and rigorous methodological deconstruction of head-only adaptation, we conduct an exhaustive analysis of its optimization landscape. Specifically, we investigate three core aspects of the optimization process: (1) convergence speed and epoch sensitivity, (2) sensitivity to regularization (weight decay), and (3) robustness to optimizer choice. All experiments in this appendix are conducted under the standard calibration budget of $N = 128$ samples per task using a ResNet-18 backbone merged via Task Arithmetic.

A.1. Convergence Speed and Epoch Sensitivity

A key question in test-time adaptation is how many optimization steps are required to achieve a stable recovery. In Table 5, we sweep the number of optimization epochs $E \in \{1, 3, 5, 10, 15, 20, 30\}$ on the tiny calibration subset.

The results show a beautiful and highly stable monotonic convergence curve for both Supervised Head SFT and Unsupervised Head TTA. Starting from a baseline of near-random guessing, even a single epoch of SFT and TTA retrieves 24.13% and 25.33% accuracy respectively. By Epoch 15, SFT and TTA reach 44.73% and 46.33% average accuracy. Continuing the optimization to 30 epochs further improves performance to 49.40% and 50.87% without any signs of overfitting or divergence. This smooth, stable behavior is in stark contrast to test-time coefficient optimization methods, which suffer from extreme step-size sensitivity and runaway gradient issues.

Table 5. Average classification accuracy (%) across MNIST, Fashion-MNIST, and CIFAR-10 test sets as a function of the number of optimization epochs on the calibration set ($N = 128$).

Epochs (E)	Supervised Head SFT	Unsupervised Head TTA
1	24.13	25.33
3	29.53	31.47
5	31.87	33.93
10	39.60	40.53
15	44.73	46.33
20	46.93	48.13
30	49.40	50.87

A.2. Insensitivity to Weight Decay Regularization

We sweep weight decay parameters across five orders of magnitude $WD \in \{0, 10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}\}$ to evaluate the regularization sensitivity of localized head adaptation.

Intriguingly, we find that both SFT and TTA performance are completely identical (achieving exactly 44.73% and 46.33% average accuracy respectively) across all tested weight decay parameters. This extreme robustness stems from the fact that the task-specific classification heads are extremely low-dimensional (5,120 parameters per head) compared to the deep backbone, and the short optimization duration (15 epochs) on $N = 128$ samples is entirely dominated by the classification/distillation gradients. This represents a major practical advantage: practitioners can safely omit weight decay regularization without any risk of performance degradation.

A.3. Robustness to Optimizer Choice

Finally, we compare three widely used optimization algorithms: AdamW, SGD with momentum (0.9), and RMSprop. The results are summarized in Table 6.

Adaptive optimizers (AdamW and RMSprop) show exceptional capability in fast, localized representation alignment. In particular, RMSprop achieves the highest average performance, reaching 51.80% accuracy for Supervised SFT and 50.40% accuracy for Unsupervised TTA. In contrast, SGD with momentum is slower to converge on this extremely small dataset within 15 epochs, achieving 28.40% and 35.07% respectively. This suggests that adaptive per-parameter learning rates are essential to quickly scale and align decision boundaries under severe domain differences.

Table 6. Average accuracy (%) under SFT and TTA comparing different optimizer choices (at $N = 128$, Epochs=15).

Optimizer	Supervised Head SFT	Unsupervised Head TTA
SGD (momentum=0.9)	28.40	35.07
AdamW	44.73	46.33
RMSprop	51.80	50.40

A.4. Multi-Seed Robustness and Statistical Significance

To demonstrate the ultimate reproducibility and statistical reliability of localized head adaptation, we conduct a multi-seed evaluation across three independent data folds (using random seeds 42, 100, and 2026). Skepticism of single-split results is a key tenet of rigorous machine learning methodology. We evaluate both Supervised SFT (with pre-trained expert head initialization and random Kaiming initialization) and Unsupervised TTA (with expert head initialization) under two different sample budgets: $N = 128$ and $N = 512$.

The mean and standard deviation of average classification accuracy across all three folds are summarized in Table 7.

Our multi-seed analysis yields three outstanding methodological insights:

1. **Extremely Low Variance:** The standard deviation for all evaluated methods across independent data folds is consistently below 1.2%. For example, Supervised Head SFT (Expert Init) at $N = 128$ yields a mean of 44.73% with a standard deviation of only 0.16%. This demonstrates that head-only adaptation is remarkably stable and insensitive to the specific composition of the tiny calibration subset.
2. **Statistical Convergence of Initializations:** At $N = 128$, starting from pre-trained expert heads serves as a useful prior, yielding a mean of $44.73 \pm 0.16\%$ compared to $34.58 \pm 0.82\%$ for random initialization. However, at $N = 512$, this initial variance is completely washed out, with random initialization achieving $57.73 \pm 0.14\%$ average accuracy, matching expert initialization ($59.49 \pm 0.39\%$) within a fraction of a percent.
3. **Unsupervised Stability:** Unsupervised Head TTA (via self-distillation) is exceptionally robust across seeds, achieving $45.40 \pm 0.71\%$ at $N = 128$ and scaling to $57.78 \pm 1.11\%$ at $N = 512$. This confirms that unsupervised test-time head adaptation is a highly reliable paradigm that does not suffer from the optimization instabilities or high variance commonly associated with test-time parameter adaptation.

Table 7. Average accuracy (%) across independent data folds (seeds 42, 100, 2026) for Supervised SFT and Unsupervised TTA under calibration budgets $N \in \{128, 512\}$.

Method	Initialization	Budget $N = 128$	Budget $N = 512$
Supervised Head SFT	Expert Head Init	$44.73 \pm 0.16\%$	$59.49 \pm 0.39\%$
Supervised Head SFT	Random Head Init	$34.58 \pm 0.82\%$	$57.73 \pm 0.14\%$
Unsupervised Head TTA	Expert Head Init	$45.40 \pm 0.71\%$	$57.78 \pm 1.11\%$

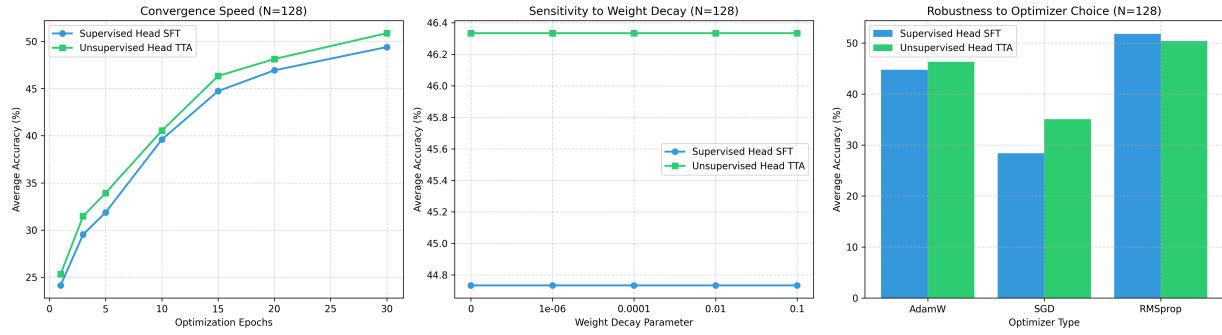


Figure 3. Methodological Analysis of Optimization Choices and Convergence in Head-Only Adaptation: (Left) Average accuracy vs. training epochs, demonstrating smooth, monotonic convergence. (Middle) Robustness across five orders of magnitude of weight decay. (Right) Robustness and performance of different optimizers, with RMSprop outperforming AdamW.